जनेप प्राधिकरण
JNPA

INDIA'S LARGEST CONTAINER PORT



CHARTING INDIA'S PORT REVOLUTION

SHRI GAURAV DAYAL ON JNPA'S GLOBAL LEAP AND THE VADHVAN VISION

As India celebrates its maritime ambitions, Jawaharlal Nehru Port Authority- India's largest container port, stands at the vanguard of transformation-rising to 23rd globally. Shri Gaurav Dayal, IAS, Chairperson of JNPA, outlines the port's ambitious journey ahead.

What are JNPA's next major capacity-expansion plans?
The recent inauguration of BMCT Phase-II has elevated JNPA's overall container-handling capacity to over 10 million TEUs, making it the largest container port in India and marking the effective utilisation of all major capacity available within the existing Nhava Sheva footprint. With brownfield expansion now fully optimised, JNPA's strategic focus is directed towards the development of Vadhvan Port, which will serve as the next-generation deep-draft gateway capable of meeting long-term national demand.

Can JNPA Lead the Next Leap in Global Port Efficiency and Competitiveness?
JNPA's rise to 23rd globally as per CPPI report 2020 - 2024 and its recognition among the top 20 most-improved ports reflect our focused drive for efficiency and performance excellence. Continual upgrades in berths, equipment, automation, and connectivity have improved turnaround time, reduced dwell time, and enabled record throughput. Beyond meeting current industry needs, we are pursuing visionary projects like the upcoming agro-processing cum storage facility and Empty Container Yard to strengthen value-added logistics. Parallelly, several other infrastructure initiatives are underway in the port to enhance the efficiency parameters at JNPA. We are also advancing sustainability, green initiatives, and smart-port technologies, ensuring we evolve as a globally competitive, future-ready port built on efficiency, innovation, and environmental responsibility.

Where does JNPA stand on adopting advanced technologies?
JNPA terminals have deployed AI-driven safety solutions that are transforming how we safeguard our people and operations. These tools don't just document incidents—they help prevent them. Key applications of AI and ML in logistics at JNPA terminals include Route Optimisation, Demand Forecasting, Inventory Management, Warehouse Automation, Predictive Maintenance, Yard Optimisation, and more. AI-Driven Solutions are transforming port and logistics operations through a suite of advanced technologies. Automated Reefer Monitoring enables real-time, intelligent tracking of refrigerated containers, ensuring cargo safety and regulatory compliance. Complementing these innovations, GPS-based vehicle monitoring and ADAS bring AI-enabled fleet management and driver safety to the forefront, creating a comprehensive ecosystem of smart, efficient, and secure port operations.

What is JNPA doing to reduce emissions and adopt renewable energy across terminals?
JNPA is accelerating its green transition with a comprehensive programme that places the port significantly ahead of national targets. JNPA has achieved 60% renewable energy share in 2025, reaching this milestone nearly five years ahead of the MIV 2030 goal, while the JNPA SEZ already functions as a 100% green-energy zone powered through in-house solar capacity and long-term green PPAs. The port is rapidly electrifying its ecosystem through EV adoption, including Zero-Emission Trucks under the ZET/Battery Swapping initiative, electric buses replacing diesel staff transport, and ongoing procurement of two hybrid (diesel–battery) green tugs. JNPA is also implementing India's first pilot project for shore-to-ship power supply for container vessels, backed by a ₹100-crore investment, and is progressing plans for a dedicated alternative-fuel bunkering facility. Terminals are contributing as well—PSA Mumbai already runs on 100% green electricity. Environmental protection infrastructure has been strengthened through an enhanced Oil Spill Response facility, Solid Waste Management and Sewage Treatment Plants, and continuous ambient air, water, and noise monitoring stations, alongside the deployment of Unmanned Solar Powered Boat to remove floating waste. The port has rejuvenated five nearby lakes, expanded solar power generation to 4.70 MWp, and increased green cover through extensive Miyawaki plantations. These initiatives are supported by a dedicated environmental laboratory, reflecting JNPA's commitment to a clean, sustainable, and green port ecosystem.

What's the progress on Vadhvan Port?
The development of Vadhvan Port project is steadily progressing. We are optimistic that the first four container terminals should be operational within five years. Vadhvan is an inflexion point not just for JNPA but for the entire Indian port sector. Near-shore reclamation and shore protection work was awarded in December 2024 to Cemindia Projects Ltd (formerly ITD Cementation). Pre-construction surveys are underway, but full site access awaits completion of land acquisition for road connectivity. We are exploring temporary leased land and alternate routes so that the near shore reclamation work can start soon. Material stockyards are only 5 km away — once access is granted, reclamation will begin immediately. The breakwater tender has received four bids. Security clearance from the Ministry of Home Affairs is under process via the Ministry of Ports, Shipping and Waterways. This should take 1–2 months, after which financial bids will be opened and the contract awarded. We will issue the work order for the breakwater within the next three months. Meanwhile, Skilling programs for various courses are actively conducted under the Vadhvan Port Skilling Program. The Board of Vadhvan Port Project Ltd (VPPL) has approved priority employment for local people based on their skills across port operations and related activities.

What progress have you made in developing the JNPA SEZ?
With 129.73 hectares allotted across 54 plots, 11 units are operational covering food processing, warehousing, and FTWZ services. EXIM activity grew to ₹16,000 crore in FY25. Based on current commitments, throughput is projected to scale significantly — 1.5 lakh TEUs in the next five years. The SEZ also incorporates a solar farm, rooftop solar, EVs and charging stations, and has entered into a 6 MW wind-power PPA, enabling the zone to operate on 100% renewable energy.



Beyond meeting current industry needs, we are pursuing visionary projects like the development of the Vadhvan Port, which is progressing steadily, ensuring we evolve as a globally competitive, futuristic port built on efficiency, innovation, and environmental responsibility.
Shri Gaurav Dayal, IAS, Chairperson, JNPA

What growth do you expect in container traffic for FY26, considering global trade volatility and changing shipping alliances?
JNPA has already become India's largest container port with capacity of over 10 million TEUs. Given the global trade environment and the ongoing reshaping of shipping alliances, JNPA remains cautiously optimistic about its throughput outlook for FY26. JNPA expects to continue its upward trajectory and maintain double-digit growth in container traffic—consistent with last year's performance—and remains hopeful of achieving 8 million TEUs.

What are the top three priorities for JNPA over the next five years?
Over the next five years, JNPA's priorities are clearly defined and aligned with both national maritime goals and the port's long-term strategic roadmap. The foremost objective is the successful progress and completion of Phase 1 of the Vadhvan Port Project, which will serve as the next major capacity engine for the Western coast and complement JNPA's existing gateway operations. In parallel, JNPA is committed to transforming itself into a 100% green and sustainable port, with a firm focus on renewable energy adoption, low-emission operations, and environmentally responsible infrastructure. The third core priority is the timely completion of ongoing, trade-enabling projects, including the Agro Processing & Storage Facility, the Shivdurg Business Complex, and other digital and physical infrastructure upgrades that strengthen operational efficiency and support the wider logistics ecosystem. These priorities collectively position JNPA for the next phase of competitive, sustainable, and service-oriented growth.